

Vibecode an ADK 2.0 Ambient Agent with Antigravity and Agents CLI

About this codelab

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1. Introduction

This lab is part of Kaggle's **5-Day AI Agents: Intensive Vibe Coding Course with Google**. You can find the additional codelabs and resources available on the [event site](#).

In this codelab you act as a **software architect**: you describe *what* you want in natural language, and **Antigravity** (Google's agentic IDE) writes and edits the code. You review, run, and verify everything **on your own machine**.

This lab is built on Google's [Agent Development Kit \(ADK\)](#), an open-source, code-first, graph-based framework for building AI agents. You'll use the **ADK 2.0 graph workflow API**, plus [agents-cli](#), the command-line toolchain for building, running, evaluating and deploying ADK agents.

The Usecase: Corporate Expense Management

Processing employee expense reports is a major administrative bottleneck. Managers are flooded with low-value, routine items (like coffee or office supplies) that could easily be automated, while high-value expenses (like flights or hardware) require careful risk reviews and manual authorization.

In this codelab, you will build an event-driven **ambient expense agent** that acts as an automated triage queue. It processes incoming expense report submissions (simulated as Pub/Sub messages) and routes them based on the transaction value:

- **Low-value expenses (Under \$100)**: Auto-approved instantly by deterministic python code (bypassing the cost and latency of LLM calls).
- **High-value expenses (\$100 or more)**: Routed through a pre-LLM security screen, analyzed for compliance risks by a Gemini LLM, and then **paused for human review**.